

Hell Cat (Bullet Hell) — Project Documentation

1. Scripts & Components

Enemies Folder

- **BulletCleanup.cs**
 - Destroys bullets when they leave the screen to prevent performance/memory issues.
 - Attached to: Bullet prefabs.
- **BulletSpawner.cs**
 - Spawns enemy bullets at intervals or in patterns.
 - Configurable fire rate, bullet prefab, and spread.
 - Attached to: Enemy GameObjects.
- **Enemy.cs**
 - Controls enemy health, movement, and collision response.
 - Calls GameManager
 - Attached to: Enemy prefabs.
- **EnemyBullet.cs**
 - Handles the movement and collision behavior of enemy bullets.
 - Attached to: Enemy bullet prefab.
- **EnemySpawner.cs**

- Spawns waves of enemies at set times or conditions.
 - References enemy prefabs.
 - Attached to: EnemySpawner GameObject in the Gameplay scene.
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Player Folder

- **PlayerController.cs**

- Handles input (via Unity's new Input System).
- Controls movement
- Attached to: Player GameObject.

- **PlayerHealth.cs**

- Tracks player HP, handles damage & death.
 - Updates UI via UIManager.
 - Attached to: Player GameObject.
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Systems Folder

- **GameManager.cs**

- Core controller of game state.
- Tracks score and game-over.
- Interfaces with UIManager to update HUD.
- Attached to: GameManager GameObject in Gameplay scene.

- **ObjectPool.cs**

- Manages reusable bullet/enemy objects.
 - Reduces instantiation cost by recycling objects.
 - Attached to: BulletPool GameObject.
 - **SceneLoader.cs**
 - Handles scene transitions (Main Menu ↔ Gameplay).
 - Attached to: SceneLoader GameObject or UI buttons.
 - **UIManager.cs**
 - Updates HUD elements (HP text and timer).
 - Controls pause and game-over panels.
 - Attached to: UIManager GameObject inside Canvas.
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Input System

- **InputSystem_Actions.inputactions**
 - Defines player input maps (move, shoot, pause).
 - Referenced by PlayerController.cs.
 - **PlayerControls.cs / PlayerControls.inputactions**
 - Auto-generated by Unity's Input System.
 - Provides script API to access inputs.
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2. GameObject Composition

Gameplay Scene (MainLevel)

- **Main Camera**
 - Orthographic camera.
 - Renders gameplay area.
- **EnemySpawner**
 - Holds EnemySpawner.cs to spawn waves.
- **GameManager**
 - Holds GameManager.cs for game state logic.
- **BulletPool**
 - Holds ObjectPool.cs for bullet reuse.
- **Canvas**
 - **UIManager** (with UIManager.cs).
 - **HUD** (child):
 - HPText (TMP_Text).
 - TimerText (TMP_Text).
 - **GameOverPanel** (child):
 - Restart Button.
 - Back to Title Button.
 - **DamageFlash**: used to flash red overlay when player is hit.
- **MainMenu (button)**
 - Provides a UI hook to return to the main menu scene.
- **EventSystem**
 - Required for UI interaction.

- **Player**

- Components:
 - SpriteRenderer, Rigidbody2D, Collider2D.
 - PlayerController.cs (movement, shooting).
 - PlayerHealth.cs (HP, death).

- **Bullet (Prefab)**

- Components: SpriteRenderer, Rigidbody2D, Collider2D.
- Scripts: BulletCleanup.cs, EnemyBullet.cs.

- **Enemy**

- Components: SpriteRenderer, Rigidbody2D, Collider2D.
- Scripts: Enemy.cs, BulletSpawner.cs.

- **BackgroundCanvas**

- Background stretched to camera size.
- Sorting Layer = BackgroundFar.

- **ScreenBounds**

- Empty with ScreenBounds.cs.
- Spawns BoxCollider2D walls at the edges of the screen.

Main Menu Scene

- **Main Camera**

- Orthographic camera.

- **Global Light 2D**
 - URP light for 2D rendering.
 - **Canvas**
 - Title Text.
 - Start Button (calls SceneLoader → MainLevel).
 - Quit Button.
 - **EventSystem**
 - Handles button interactions.
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3. Scene Descriptions

Main Menu Scene

- Simple UI with background, title, and buttons.
- Start Game loads the MainLevel.
- Quit closes the application.

Gameplay Scene (MainLevel)

- Player spawns in the center.
- Enemies spawn from EnemySpawner.
- Bullets are managed by a pooling system.
- UI displays HP and Timer.
- The GameOver panel appears when the player dies.
- The restart button reloads MainLevel.

- Back to Title button loads Main Menu.
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4. Expected Interactions

- **Player ↔ ScreenBounds**: Player cannot leave the screen.
 - **Player ↔ EnemyBullets**: Player takes damage.
 - **GameManager ↔ UIManager**: Updates HUD and game-over state.
 - **Buttons ↔ SceneLoader**: Change scenes between Menu and Gameplay.
 - **ObjectPool ↔ Bullets/Enemies**: Efficiently recycles objects.
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5. External Resources

- **Unity Input System**: for handling movement/shooting input.
- **Tutorials/Guides used**:
 - <https://www.youtube.com/watch?v=-GWjA6dixV4>
 - Helped create a main menu and game menu
 - <https://www.youtube.com/watch?v=XtQMytORBmM>
 - Helped with the overall functionality of the game
- **ChatGPT 5.0**:
 - 9/8: I am creating a Bullet Hell like game in Unity. Can you give me a plan/layout of what I would need to accomplish this goal?
 - 9/14: so i restarted unity and now my main menu scene is gone how do i get it back
 - 9/24: Im trying to create a background/environment for my game, can you show me how I would do that
- **Game art used**:
 - <https://opengameart.org/content/dog-spritesheets>
 - Enemy sprite
 - <https://opengameart.org/content/missiles-bullets-and-bomb-set>
 - Bullet sprite
 - <https://opengameart.org/content/catgolden-cat>
 - Player sprite

- <https://opengameart.org/content/tower-defense-grass-background>
 - Background image